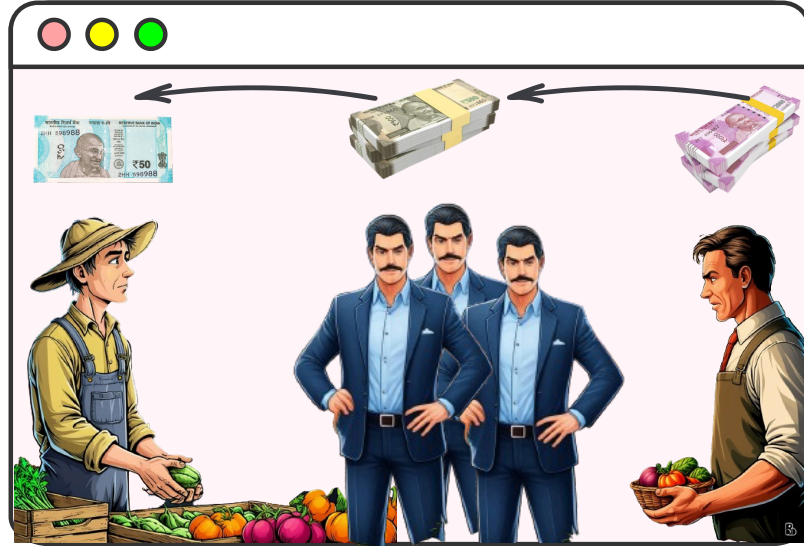




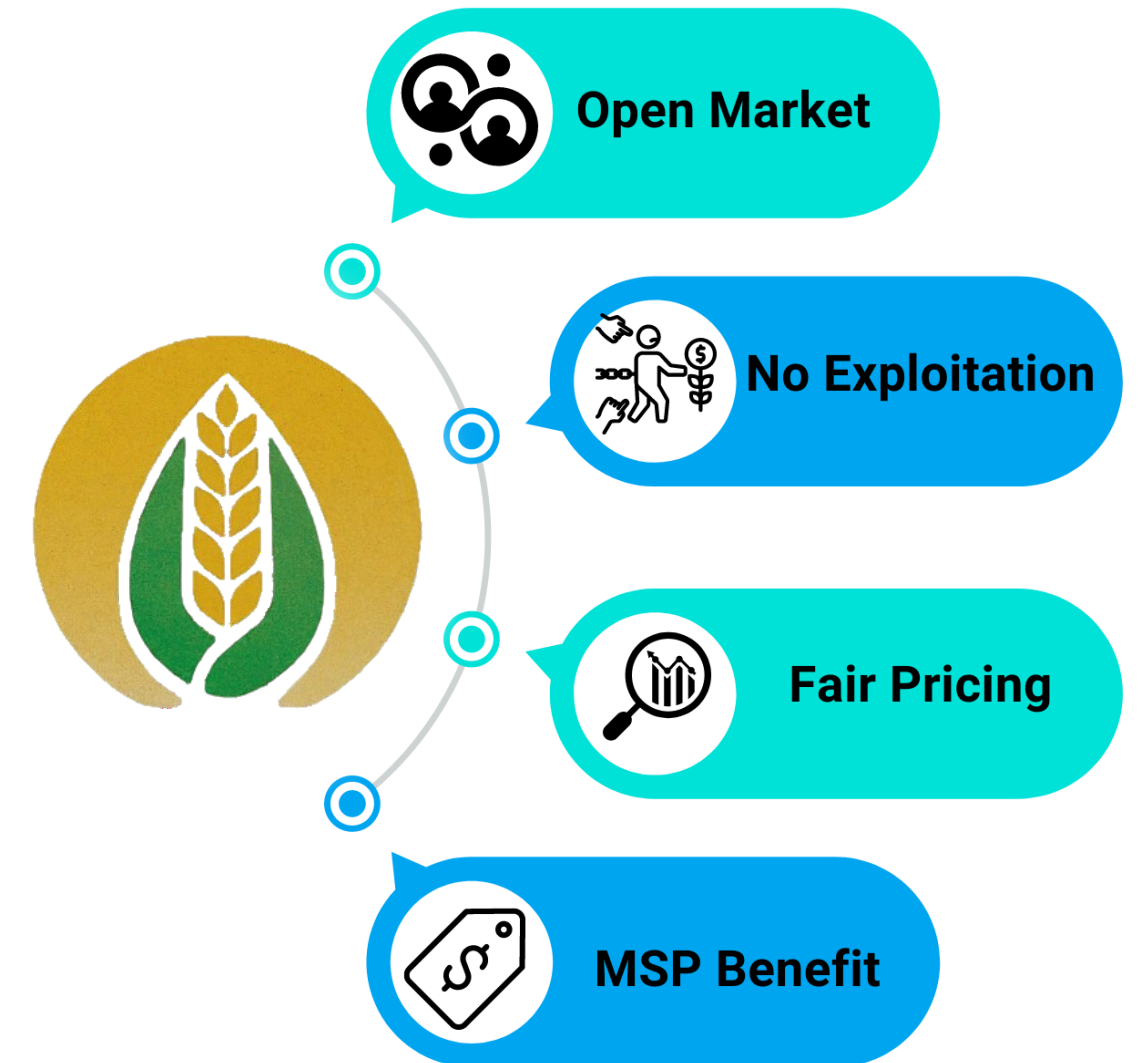
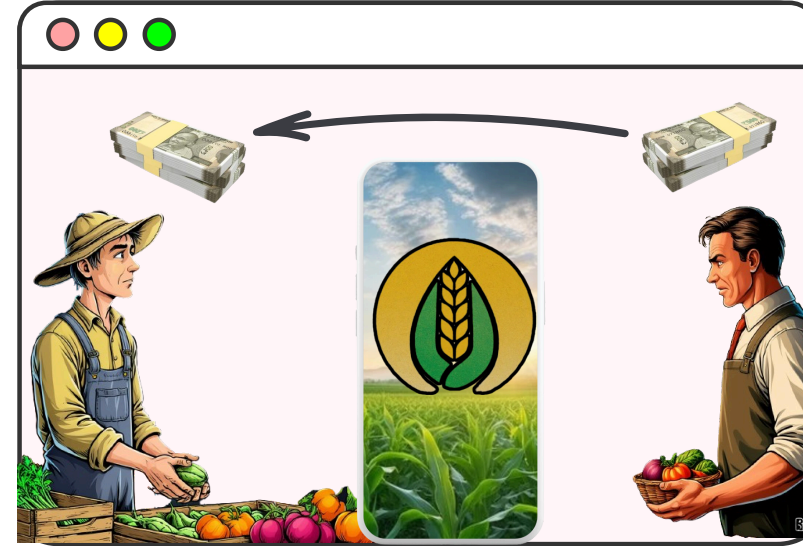
TITLE PAGE

- **Problem Statement ID – 1637**
- **Problem Statement Title – Mobile App for Direct Market Access for Farmers**
- **Theme – Agriculture , Food Tech & Rural Development**
- **PS Category - Software**
- **TeamID- 14306**
- **Team Name - CropCoder**

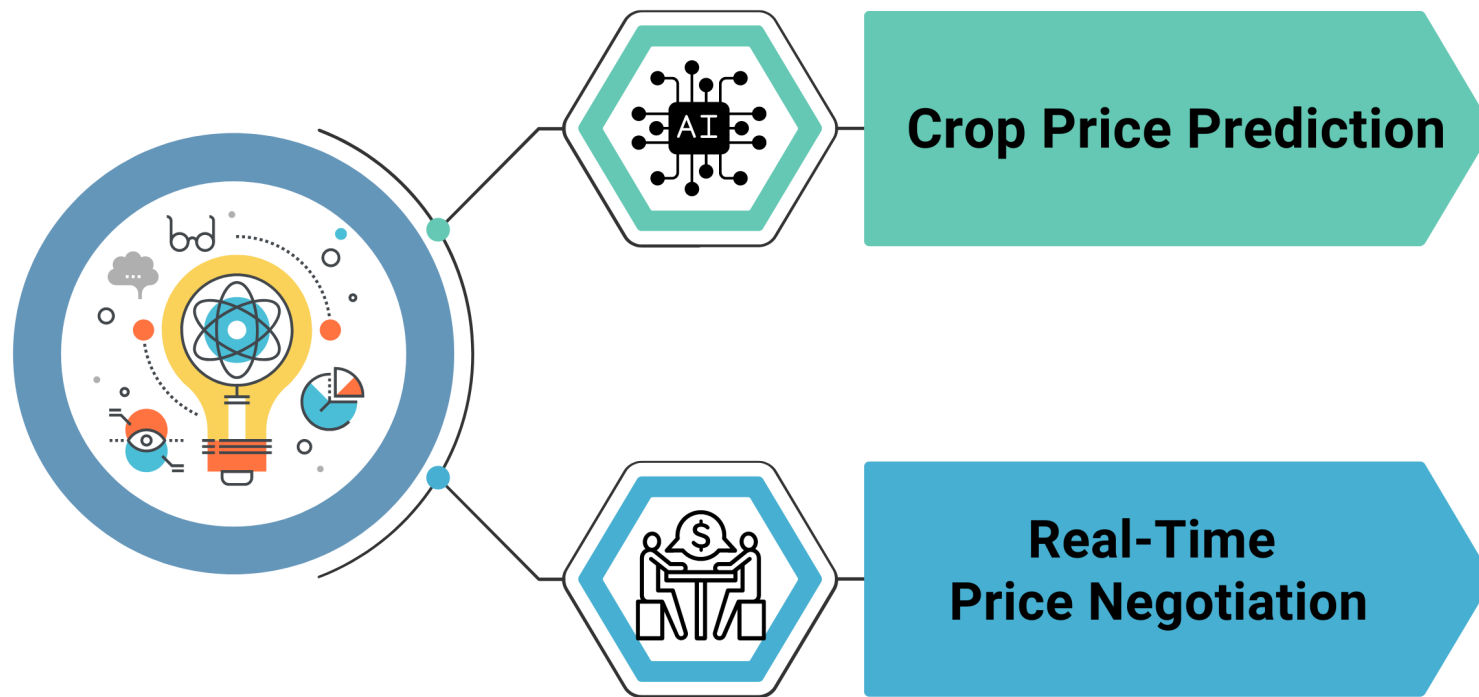
Proposed Solution :



- **Directly connects** farmers with consumers and retailers.
- **Prevents exploitation** concerns.
- **Empowers farmers** to sell at competitive prices.
- **Guarantees MSP** benchmarking.



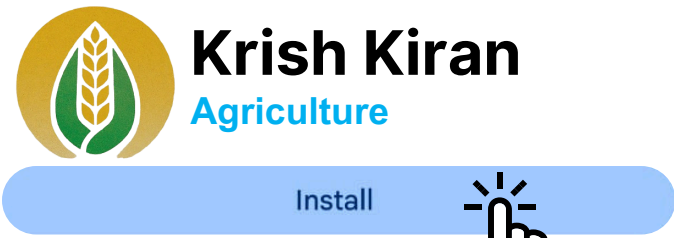
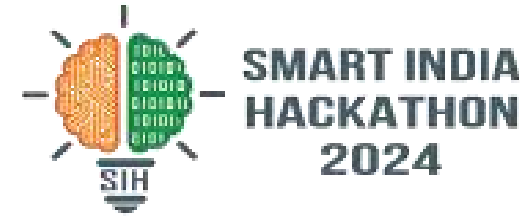
Innovation and Uniqueness of the Solution :



- AI-driven crop price forecasting helps farmers sell at the right time, ensuring profitable returns in line with the Essential Commodities Act 2020.
- Facilitates direct negotiation between buyers and sellers for **transparent** and **fair market rates**.



TECHNICAL APPROACH



Username

Password

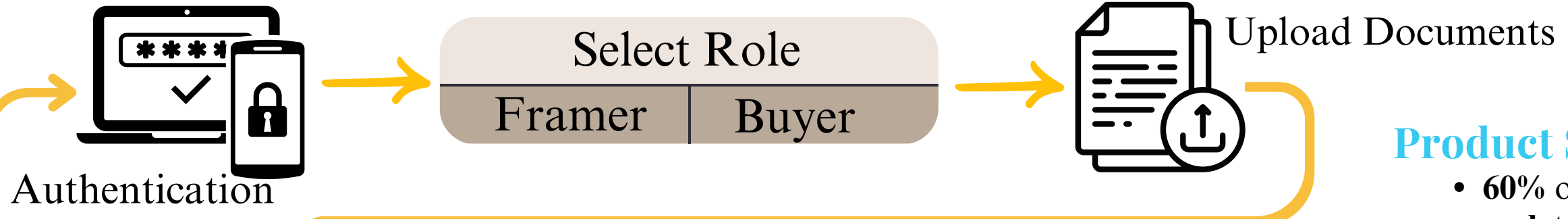
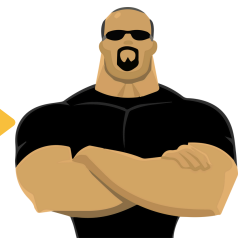
LOGIN

Explore Interface

Find App Useful

YES

NO



Real Time Market Insights

FOR SALE

List Product For Sale

Apply For Quality Check

MSP Benchmarking

Negotiate With Buyer

Customer Support

Farmer

Search Product

Price Comparison

Add to Cart

Negotiate With Farmer

Track Order

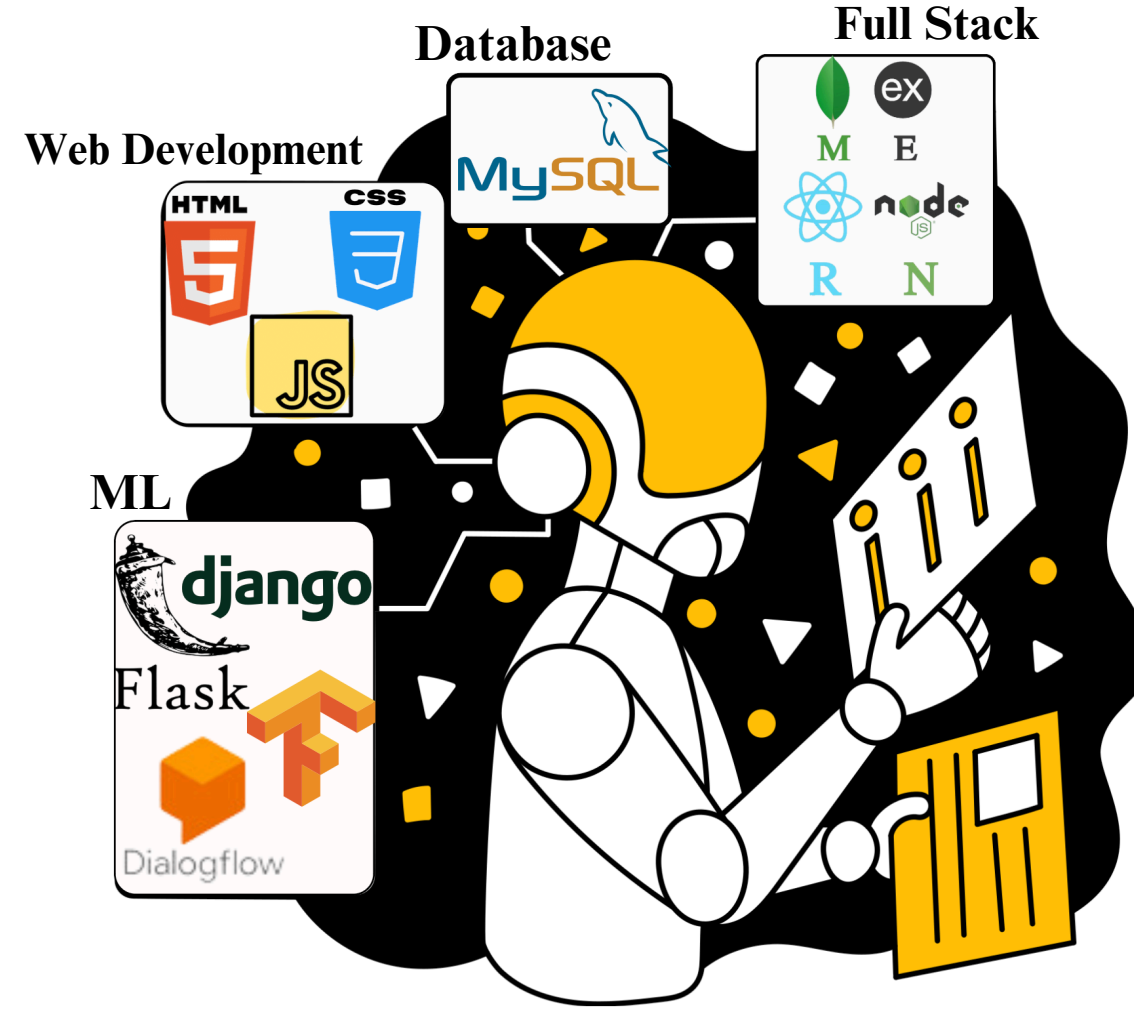
Customer Support

Buyer

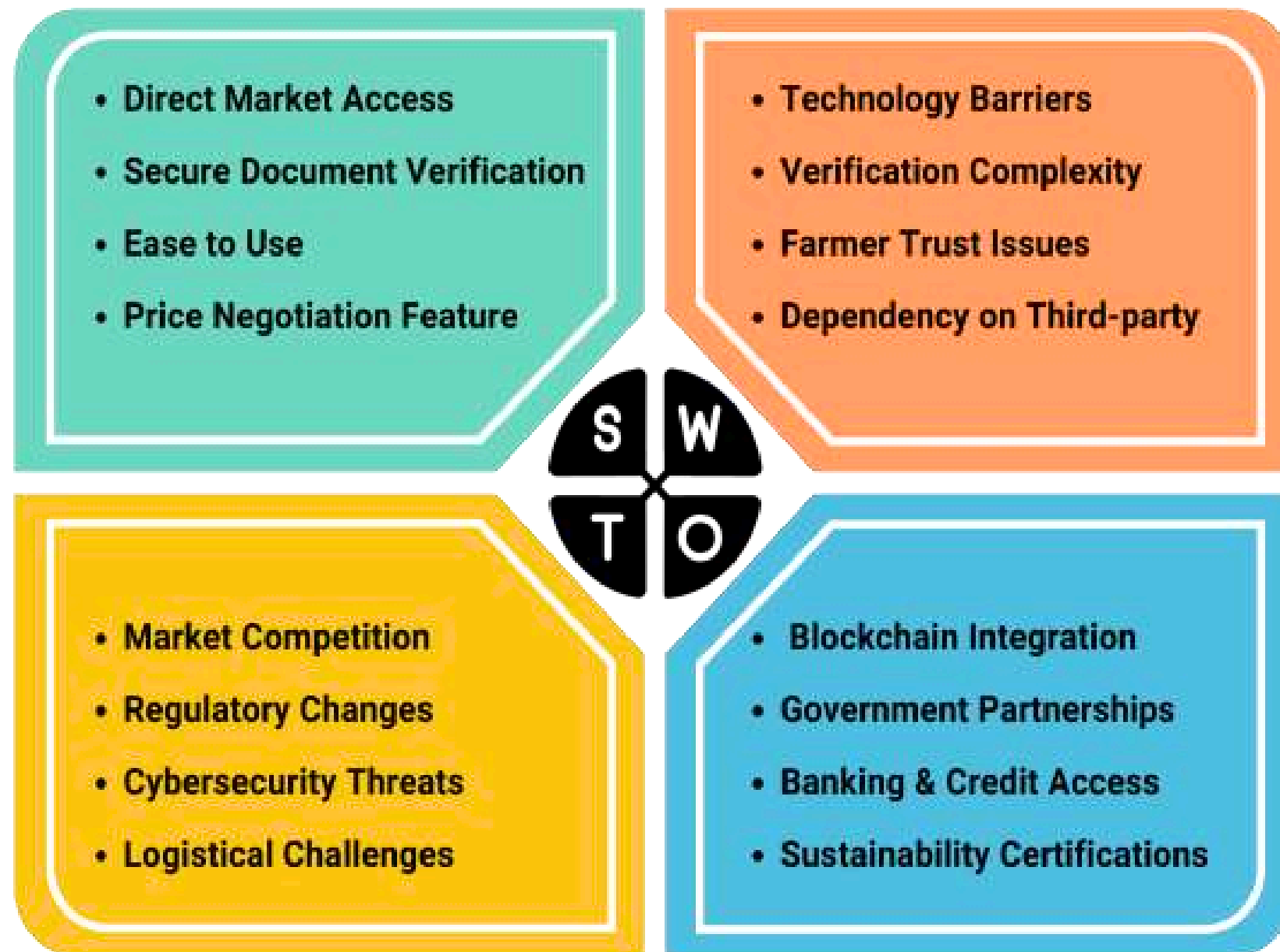
Delivery Confirmation

Product Status :

- 60% of the product built is complete.
- Remaining development is in progress.
- Machine Learning (ML) is deployed in Backend.
- Testing and Validation are underway.



SWOT Analysis :



Market Feasibility :

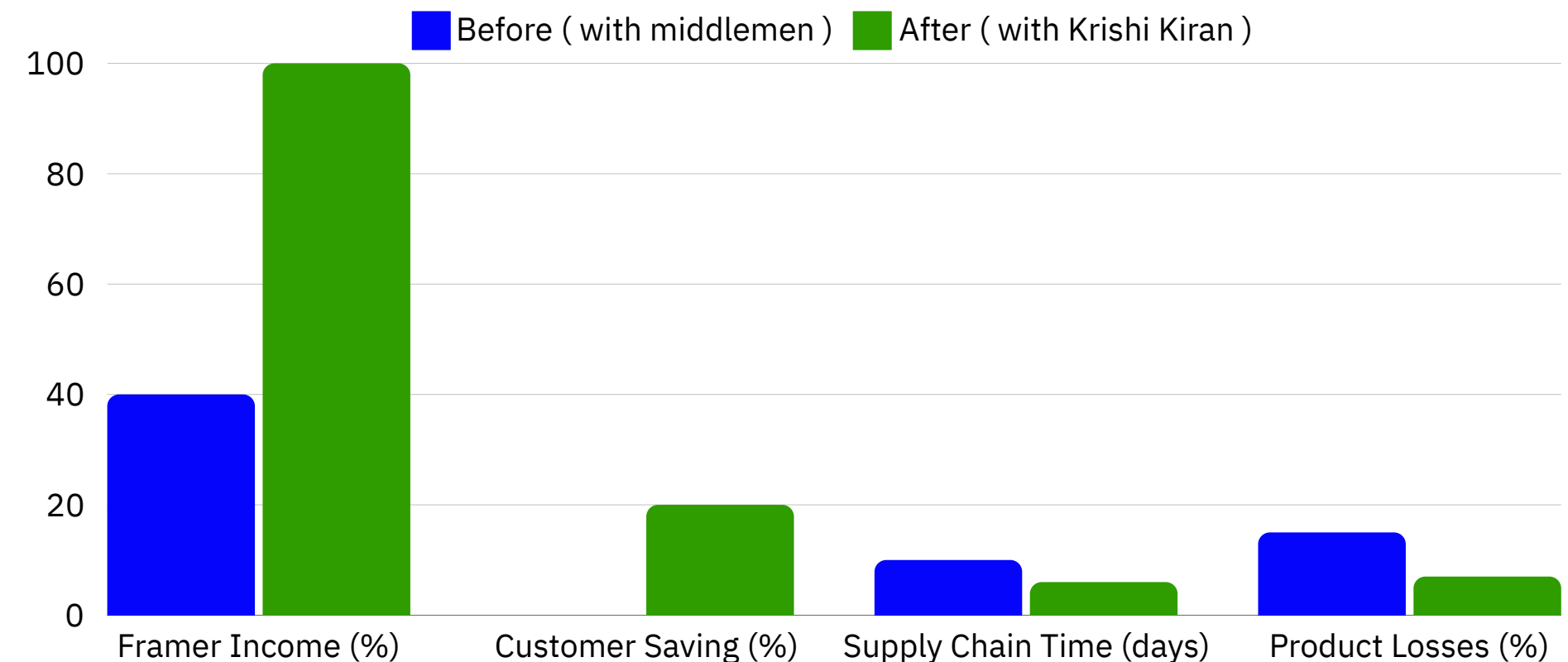
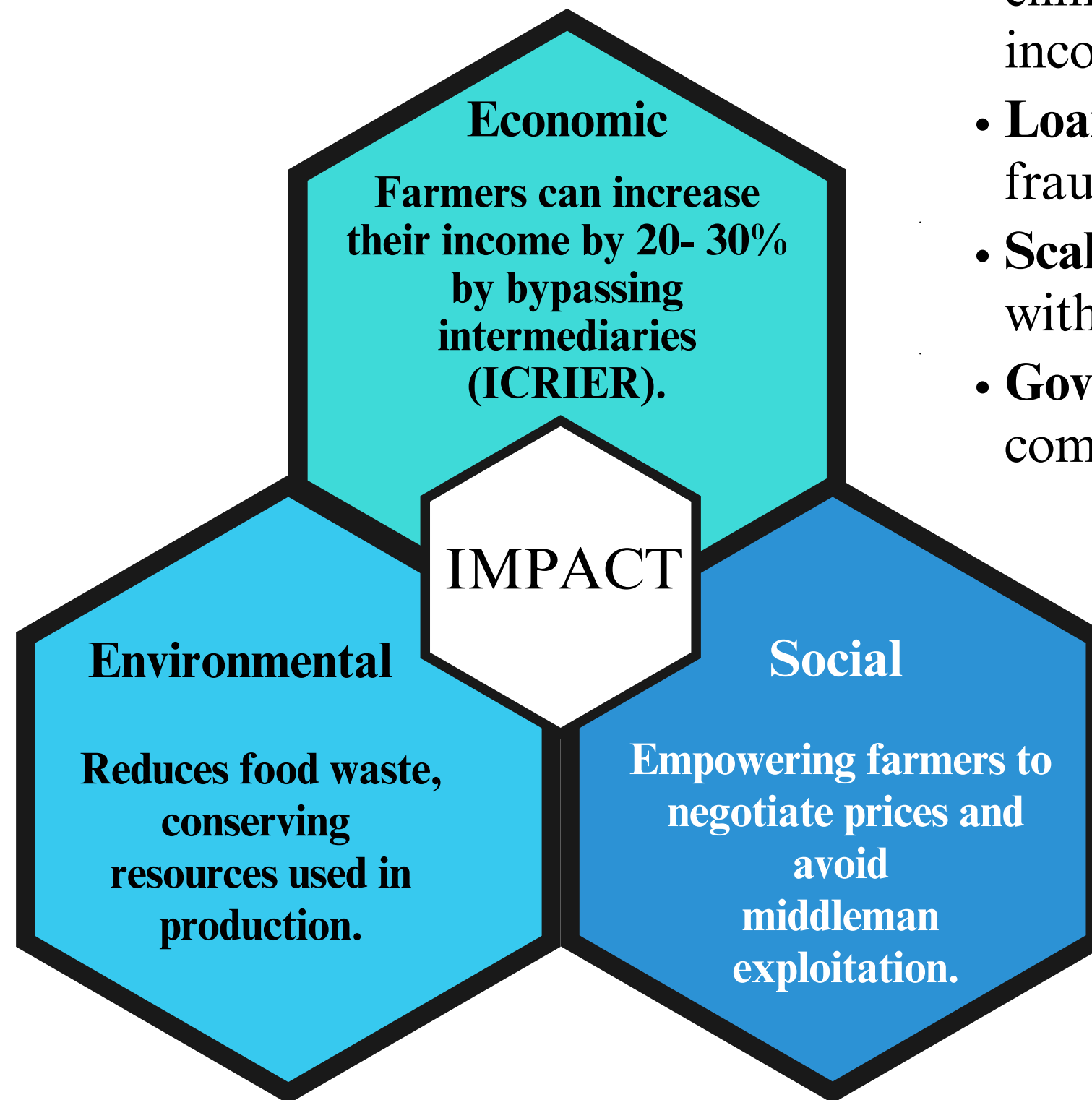
- With over **60% of farmers** lacking access to digital marketplaces.
- The global Agri-tech market is projected to reach **\$22.5 billion by 2025**.
- Expanding to additional commodities and offering sustainability certifications can attract new users.

Financial Feasibility :

- Revenue will be generated through **transaction fees (1-2%)**, a **subscription model** for premium features, and **partnerships** with Agri-tech
- The project may seek **government grants** or **investor funding** to cover scaling costs, marketing, and operational expenses like server maintenance and customer support.

Benefits of Krishi Kiran:

- **Empowerment and Financial Inclusion:** Can Increase farmer incomes by eliminating middlemen, aligning with the government's goal of doubling farmer incomes.
- **Loan Fraud:** Farmers may falsify crop damage reports to secure loans fraudulently.
- **Scalability Across India:** Adaptable to different regions and crops, aligning with "Digital India" initiatives.
- **Government Alignment:** Supports the objectives of the **2020 farm bills** and complements existing government schemes like **e-NAM**.





RESEARCH AND REFERENCES



Research:

- **Market Research**
 - **Farmer Income:** Farmers lose 20-30% of income due to middlemen (ICRIER).
 - **Consumer Demand:** Urban consumers seek fresh, organic farm-to-table produce.
- **Technology Research**
 - **Blockchain:** Ensures transparency and trust in transactions.
 - **Security:** Multi-factor authentication to prevent fraud and fake profiles.
- **Supply Chain Efficiency**
 - **Post-Harvest Losses:** Improved logistics can reduce 5-15% losses (PIB).
 - **Direct Access:** Boosts farmer income by 20-30% through transparent sales.
- **Regulatory Framework**
 - **2020 Farm Bills:** Supports farmer market access and fair pricing, aligning with government reforms.
- **Competitor Insights**
 - **Existing Platforms:** e-NAM, DeHaat offer access but lack real-time negotiations and blockchain transparency.

References:

- **Goyal, Aparajita.** "Information, Direct Access to Farmers, and RuralMarket Performance in Central India." Policy Research Working Paper 5315, The World Bank, Development Research Group, Agriculture and Rural Development Team, May 2010. https://www.researchgate.net/publication/40783722_information_Direct_Access_to_Farmers_and_Rural_Market_Performance_in_Central_India
- **Thapaswini, G., & Gunasekaran, M. (2022).** A methodology for crop price prediction using machine learning. *2022 IEEE 2nd International Conference on Mobile Networks and Wireless Communications (ICMNWC)*, 1-6. <https://doi.org/10.1109/ICMNWC56175.2022.10031852>
- **Gao, X., Chen, S., Zheng, Y., & Hao, J. (2021).** A deep reinforcement learning-based agent for negotiation with multiple communication channels. In 2021 IEEE 33rd International Conference on Tools with Artificial Intelligence (ICTAI) (pp. 868-872). IEEE. <https://doi.org/10.1109/ICTAI52525.2021.00139>